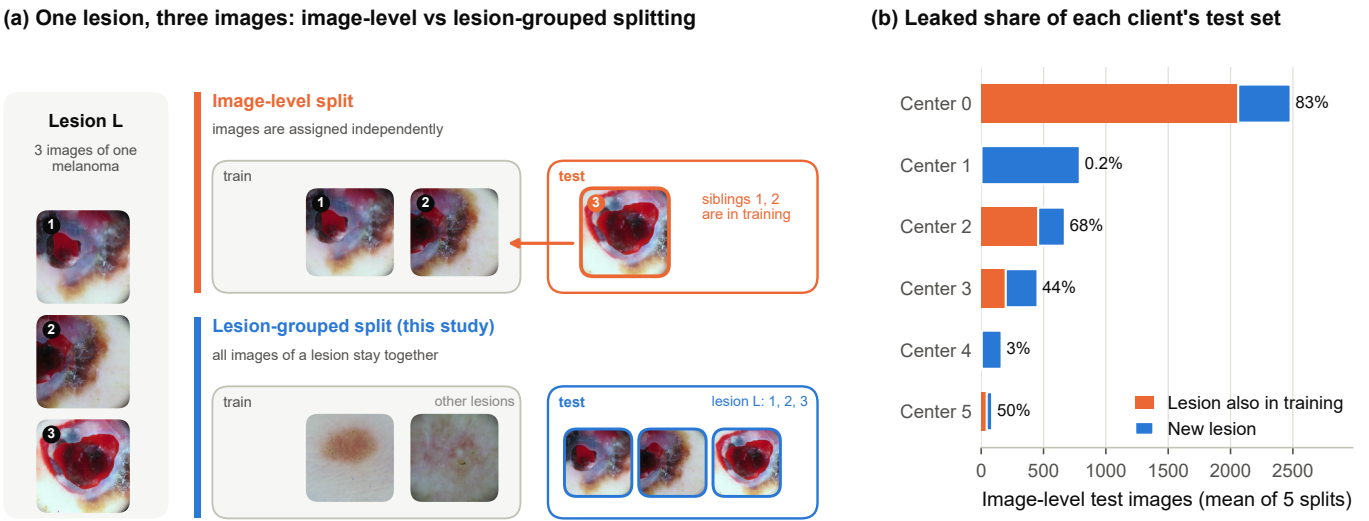

(a) One lesion, three images: image-level vs lesion-grouped splitting
Lesion L
3 images of one melanoma
1
2
3
Image-level split
images are assigned independently
train
1
2
test
3
siblings 1, 2 are in training
Lesion-grouped split (this study)
all images of a lesion stay together
train
other lesions
test
lesion L: 1, 2, 3
(b) Leaked share of each client's test set
Center 0
83%
Center 1
0.2%
Center 2
68%
Center 3
44%
Center 4
3%
Center 5
50%
Lesion also in training
New lesion
0
500
1000
1500
2000
2500
Image-level test images (mean of 5 splits)